# Loan Default Prediction

## 1. Introduction:

Building and assessing machine learning models to forecast loan defaults using historical loan data is the main goal of this project. Finding possible loan defaults is intended to assist lenders in managing risk, making better decisions, and possibly optimizing loan portfolios. This project intends to give lenders useful advice by examining a number of variables related to loan applications and repayment patterns in order to shed light on the traits of borrowers who are most likely to default.

## 2. Dataset Exploration:

**Data Overview:** A summary of the data, including data types, non-null counts, and descriptive statistics for numerical columns, was obtained through initial inspection using head(), tail(), info(), and describe().

**Data Shape and Columns:** To comprehend the dimensions and features available, the dataset's shape (df.shape) and column names (df.columns) were analyzed.

**Missing Values and Uniqueness:** Columns with unique values were found using isnull() and nunique().The existence and number of missing values were disclosed by sum(). There were no missing values discovered.

**Value Duplicity:** duplicated().The dataset's lack of duplicate rows was verified by sum().

3. **Exploratory Data Analysis:**

**Numerical Feature Distribution:** To see the distributions of numerical features, histograms (df.hist()) were created. To find possible outliers and comprehend the distribution of the data for every numerical column, box plots were also made.

**Categorical Feature Distribution:** The frequency distribution of categorical features was visualized using count plots (sns.countplot()), which shed light on the various categories and their counts.

**Pair Plot Relationship:** To illustrate the linear relationships between numerical features, a correlation matrix (sns.heatmap()) was calculated and displayed. This made it easier to spot possible multicollinearity and comprehend the relationships between numerical variables.

**<u>Correlation Matrix Relationship:</u>** For a thorough understanding of possible correlations and patterns, a pair plot (sns.pairplot()) was created to show the pairwise relationships between all numerical features and the target variable.

4. **<u>Model Training:</u>**

LightGBM, Support Vector Machine (SVM), and Random Forest were the three machine learning models that were trained and evaluated for the project's loan default prediction.

**<u>Data Preprocessing:</u>** LabelEncoder was used to convert the categorical "purpose" column into a numerical format.

The target variable (y) and features (X) were kept apart.
To preserve the percentage of the target variable in both sets, the dataset was divided into training and testing sets using train_test_split, with a test size of 20% and stratified sampling.

**Managing Class Imbalance:** There were noticeably more non-defaulting loans in the dataset than defaulting loans, indicating a class imbalance. To remedy this, synthetic samples for the minority class (defaulting loans) were created using the Synthetic Minority Over-sampling Technique (SMOTE) on the training data, producing a balanced

**Model Names:**

- ✓ **LightGBM Classifier**: An lgb.LGBMClassifier was trained on the balanced training data.

- ✓ **SVM Classifier:** An SVC model was trained on the balanced training data.
- ✓ **Random Forest Classifier:** A RandomForestClassifier was trained on the balanced training data.

5. **Model Evaluation:**

Several metrics were used to assess the trained models on the dataset:

**Precision:** It is the percentage of correctly predicted positive cases among all positive cases.

**Recall:** The percentage of accurately predicted positive instances among all actual positive instances.

**F1-Score:** A balanced metric derived from the harmonic mean of precision and recall.

**Classification Report:** A thorough report that breaks down each class's precision, recall, F1-score, and support.

**Confusion Matrix:** A table that displays the numbers of true positive, true negative, false positive, and false negative predictions.

## 6. Conclusion:

This project effectively created and assessed machine learning models for predicting loan default. The models offer important insights into loan default risk by utilizing data analysis and addressing class imbalance. Each model's strengths are highlighted by the evaluation metrics, and LightGBM performs well overall, especially when it comes to spotting possible defaults. The suggestions made are meant to help lenders use these models to enhance their risk management and decision-making procedures.

## 7. Future Works:

Future research could focus on the following areas:

**Feature Engineering:** Investigating the development of new features from preexisting ones that could enhance model performance.

**Hyperparameter Tuning:** Improving the selected models' hyperparameters to increase their capacity for prediction.

**Examining Other Models:** Examining how well alternative machine learning models perform when used for binary classification tasks.

**Examining the Influence of External Factors:** Taking into account market trends or external economic factors that could affect loan defaults.

**Creating a Production System:** Including the learned model in a system that evaluates loan applications in real time.